

Meghaa K

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Education

M.E Artificial Intelligence and Machine Learning
(8.68 till 1st semester)

Manipal School of Information
Sciences (2025 to 2027)

B.E in Computer Science and Engineering
(9.28)

S.E.A College of Engineering and Technology,
Bangalore

Class 12
(87.33)

St Annes Girls Pre University, Bangalore

SSLC / Class 10
(92%)

St Annes Girls High School, Bangalore

Experience

- **Open-Source Contributor, GirlScript Summer of Code (GSSoC)** | Oct 2024 to Nov 2024

Contributed to multiple open-source projects, earned 7 badges and achieved **Global Rank 457** among thousands of participants.

- **Team Leader and Finalist, Web3Conf India** | Sep 2024 to Dec 2024

Worked on the project of Encrypted Cryptographic QR Code Generator Project, improving data confidentiality in supply-chain use cases, Reached finals among national-level participants.

- **Member, Women Techmakers** | Jul 2024 to Present

Active member of Women Techmakers community and participated in tech events, workshops, and networking activities.

Technical Skills

Languages: Python, C++

Machine Learning & NLP: Retrieval Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Embeddings

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, LangChain, HuggingFace Transformers, Sentence-Transformers, FAISS

Data & Databases: SQL (basics), Vector Databases (FAISS)

Visualization & UI: Streamlit, Power BI, Matplotlib

Tools & Platforms: Git, GitHub, Visual Studio Code, Jupyter Notebook

Projects

Document-Grounded RAG LLM Assistant for Semiconductor Knowledge Systems:

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for querying private semiconductor manuals and enterprise documents using HuggingFace embeddings, FAISS vector search, and Llama3 via Ollama,
- Built offline ingestion and online inference pipelines with chunking, semantic indexing, and top-k retrieval to reduce hallucination and improve domain accuracy.
- Integrated Streamlit frontend with LangChain RetrievalQA to deliver grounded answers with source citations.
- Enabled scalable enterprise-style document QA for architecture, cache, memory, and ISA documentation.

Diabetic Retinopathy Detection using Explainable Artificial Intelligence:

- Applied Grad-CAM to visualize important regions in retinal images, making the model explainable and trustworthy and focused on improving medical decision support using AI interpretability techniques.

Mitosis Detection using YOLO and DINO:

- Built an object detection model to identify mitosis cells in microscopic images using **YOLO v5** and **DINO** architectures for accurate detection with accuracy of 91%.

Encrypted Cryptographic QR Code Generation for carbon footprint reduction and Eco-Friendly Supply Chains

- Python-based application to generate AES-encrypted QR codes for secure data sharing, supports paperless and secure digital documentation for real-world applications.

Certifications

- Machine Learning for All, University of London
- AI Agents Course, Hugging Face
- LLM Course, Hugging Face

Publications

Development of a Lightweight Employee Information System using Java and MySQL: A Backend Optimization Perspective (Journal: International Scientific Journal of Engineering and Management)

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